# ADITYA MAHAMUNI

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LinkedIn | Portfolio | GitHub

Machine Learning Engineer focused on production GenAI, agentic systems, evaluation, and controls. Experienced in agent frameworks, backend services, synthetic data, adversarial evaluation, RAG/MCP tooling, and AWS Bedrock/SageMaker. Strong background in Python, distributed systems, CI/CD, and cross-team delivery.

## TECHNICAL SKILLS

**Programming:** Python, C++, SQL

**Software engineering:** Object-oriented programming, design patterns, FastAPI, REST APIs, event-driven architecture, Kafka, PostgreSQL, Pydantic, JWT, distributed systems

**AI / LLM engineering:** LLM applications, agentic AI, RAG, prompt and context engineering, structured outputs, tool-using agents, synthetic data, evaluation and controls, adversarial evaluation, prompt-injection testing, MCP

**Agent frameworks:** LangGraph, OpenAI Agents SDK, CrewAI, PydanticAI, smolagents, AutoGen

**Tools and cloud:** Linux, Docker, Jenkins, GitLab CI, Git, JIRA, Kibana, Grafana, Agile (Scrum), AWS Bedrock, AWS SageMaker

**Machine learning:** NLP, Transformers, PyTorch, TensorFlow, PEFT / LoRA concepts, fine-tuning strategy, model evaluation and deployment

## EXPERIENCE

**NatWest, London, UK - Machine Learning Engineer**
Dec 2024 - Present

- Built and evaluated agent framework prototypes using LangGraph, CrewAI, OpenAI Agents SDK, PydanticAI, smolagents, and AutoGen; developed reusable patterns for routing, tool use, handoffs, and multi-agent orchestration.
- Built a production FastAPI microservice to orchestrate tool-using AI agents, with a focus on reliable service integration and maintainable software design.
- Designed and built a reusable synthetic-data generation framework with a domain-agnostic core and pluggable adapters, using structured LLM outputs and validation to support development and evaluation.
- Contributed to the design and development of a GenAI evaluation and controls framework to support consistent assessment of LLM applications and agentic systems.
- Designed adversarial and prompt-injection evaluations using synthetic data, and used targeted scenarios to assess model quality and refine prompts.
- Developed reusable coding-agent skills for repeatable, agent-assisted engineering tasks.
- Led cross-team delivery, owned technical design decisions, and communicated progress and outcomes to stakeholders and leadership.

**NVIDIA Corporation, Pune, India - Software Developer**
July 2019 - Aug 2023

- Designed a scalable Python framework and training scenarios for autonomous vehicle simulation.
- Optimized CI/CD pipelines using Python, GitLab CI, and Docker to improve deployment efficiency and scenario updates.
- Created automated testing frameworks with Jenkins and Python to reduce manual effort and shorten testing cycles.
- Developed ML tools and CI/CD pipelines using NLP, PyTorch, and TensorFlow.
- Built a Python and Linux diagnostic tool to improve platform reliability.
- Led technical projects and collaborated with cross-functional teams to deliver against project requirements.

## EDUCATION

**University of Warwick**, Coventry, UK | Sept 2023 - Sept 2024
MSc Computer Science - Degree classification: 2:1

**Government College of Engineering**, Aurangabad, India | July 2015 - May 2019
Bachelor of Engineering, Information Technology - CGPA: 7.448/10.0

## PROJECTS

### Event Extraction from Financial Articles Using Natural Language Processing

MSc dissertation. Developed NLP techniques to identify and classify financial events using zero-shot and few-shot LLM learning, generative models, and fine-tuning.

### A Clustering and Regression Analysis of Financial Health and Stock Performance

Built a Python-based financial analysis project using clustering and regression to explore company financial health, market patterns, and stock performance.

### Prediction of Protein Expression

Used image analysis to predict protein expression in biological tissues, exploring PCA, GLCM, transfer learning, OLS, and SVR. Achieved an RMSE of 0.677 using a CNN architecture.

## CERTIFICATION

Neural Networks and Deep Learning - DeepLearning.AI | Sept 2020